

Khwaja Yunus Ali University

Founder: Dr. M. M. Amjad Hussain

School of Science & Engineering

Department of CSE & EEE

Spring - 2024

Course Outline: Physics – II

Part – A:

1. **Course Code:** CSE 0533 1201 (CSE), PHY 0533 1201 (EEE)
2. **Course Title:** Physics - II
3. **Course Type:** Related
4. **Level:** 1.2 (1st Year 2nd Semester)
5. **Academic Session:** Spring - 2024
6. **Course Instructor:** Md. Matiur Rahman
7. **Prerequisites:** N/A
8. **Credit Value:** 3.00
9. **Contact hours/week:** 3.00 hours
10. **Total Marks:** 100
11. **Rationale of the Course:** This course is to give students a broad understanding of physics and a foundation in problem solving and critical thinking. Topics covered in this course include Thermodynamics, Modern Physics and Optics.

12. Course Objectives

The principles of physics are the basis of all modern technology, and understanding the concepts of physics and knowing how to solve physics problems is a key indicator of success in advanced study in all technical fields, including biology, medicine, and the health sciences. The purpose this course is to give students a broad understanding of physics and a foundation in problem solving and critical thinking. Topics covered in this course include Thermodynamics, Modern Physics and Optics.

13. Course Learning Outcomes (CLOs) and its mapping with PLOs:

CLOs	After successful completing this course, students will be able to.	PLO Mapping
CLO1	Define the basic terms regarding thermodynamics, modern physics and optics.	PLO1
CLO2	Apply the critical thinking and problem-solving skills to solve the variety of basic problems in Heat, modern physics and optics.	PLO2 & PLO9
CLO3	Demonstrate the fundamental physics principles using modern tools.	PLO5
CLO4	Analyze and describe the natural phenomenon which are observed in practical situation in everyday life.	PLO4 & PLO7
CLO5	Create ideas to do the research work for the advancement of technological world.	PLO3 & PLO11,12
CLO6	Apply ethical responsibilities and professional conduct.	PLO8

Part – B:

14. Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week	Topic	Teaching Strategy	Assessment Technique	Alignment with CLOs
01	Principle of temperature measurements: Platinum resistance thermometer, Thermo-electric thermometer, Pyrometer	Interactive discussion / Group activity / Discussion / Presentation / Laboratory Work	Quiz / Class test / Assignment / Presentation / Midterm / Final exam.	CLO1
02	Kinetic theory of gases: Maxwell's distribution of molecular speeds, Mean free path, Equipartition of energy, Brownian motion, van der Waal's equation of state			CLO2 & CLO3
03	Review of the First law of thermodynamics and its application			CLO2 & CLO3
04	Reversible & irreversible processes, Second law of thermodynamics, Carnot; Efficiency of heat engines, Carnot theorem, Entropy and Disorder			CLO1, CLO2 & CLO3
05	Thermodynamic Functions, Maxwell relations, Clausius-Clapeyron equation, Gibbs phase rule, Third law of thermodynamics			CLO2, CLO3 & CLO5
06	Interpretation of Bohr's postulates, Properties of nucleus, nuclear reactions, Fission. Fusion, Chain reaction, nuclear reactor			CLO1 & CLO3
07	Wave particle duality, Compton effect, Photo-electric effect, Quantum theory of radiation			CLO2 & CLO3
08	Radioactive disintegration			CLO2, CLO3 & CLO4
09	Michelson-Morley's experiment, Frame of references, Galilean transformation, Lorentz transformations			CLO2, CLO3 & CLO4
10	Special theory of relativity & its consequences			CLO2 & CLO4
11	Interference of light, Young's double slit experiment, Displacements of fringes & its uses. Fresnel Bi-prism, Interference at wedge shaped films, Newton's rings, Interferometers			CLO1 & CLO2
12	Diffraction of light: Fresnel and Fraunhofer diffraction. Diffraction by single slit. Diffraction from a circular aperture, Resolving power of optical instruments, Diffraction at double slit & N-slits-diffraction grating.			CLO2 & CLO3
13	Polarization: Production & analysis of polarized light, Brewster's law, Malus law, Polarization by double refraction. Retardation plates.			CLO2, CLO3 & CLO4
14	Nicol prism. Optical activity. Polarimeters, Polaroid.			CLO4, CLO5 & CLO6

Part – C:

15. Assessment and Evaluation

1. Assessment and Evaluation

1. Assessment Strategy

Students are evaluated out of total 100 Marks. There are different types of assessment tools. In the **Mid Term** examination, total time is **1.5 hours** where the marks is **30**. In the **Final Examination**, total marks is **40** and allotted time is **2 Hours**. The detail with **CLOs-Assessment Mapping** is given below.

CLOs	Assessment Tools (100%)					
	20%			30%	40%	10%
	Class Test	Quizzes	Assignments	Midterm Exam.	Final Exam.	Attendance
CLO1	√	√		√	√	√
CLO2	√	√		√	√	√
CLO3			√	√	√	√

CLO4			√	√	√	√
CLO5				√	√	√
CLO6			√			√

2. Evaluation System (Grading System)

Numerical Number	Letter Grade		Grade Point
80 % and above	A+	(A Plus)	4.00
75 % to less than 80 %	A	(A Regular)	3.75
70 % to less than 75 %	A-	(A Minus)	3.50
65 % to less than 70 %	B+	(B Plus)	3.25
60 % to less than 65 %	B	(B Regular)	3.00
55 % to less than 60 %	B-	(B Minus)	2.75
50 % to less than 55 %	C+	(C Plus)	2.50
45 % to less than 50 %	C	(C Regular)	2.25
40 % to less than 45 %	D		2.00
Less than 40 %	F		0.00

2. Marks Distribution

a) Continuous assessment (30 Marks)

Bloom's Category Marks (out of 30)	Class Tests (07)	Assignments (07)	Quizzes (06)	Curricular / Co- Curricular Activities (10)
Remember	02			Attendance
Understand	02			
Apply	03			
Analyze		02		
Evaluate		02	03	
Create		03	03	
Total	07	07	06	10

b) Summative: Semester (Midterm 30 marks and Final exam 40 marks) Total: 70 Marks.

Bloom's Category	Midterm (30)	Final (40)	Others Evaluation
Remember	5	8	10
Understand	5	8	
Apply	5	6	
Analyze	5	6	10
Evaluate	5	6	
Create	5	6	
Total	30	40	

3. Make up procedures:

For weak students, we take extra class before their midterm and final examinations. Beside this, we counseling with them and solve their problem as soon as possible.

Part – D:

16. Learning Materials:

1) Recommended Reading

- Physics for Engineers (Part – 1 & Part – 2) – Gias Uddin Ahmed
- Halliday Resnick - Fundamental of Physics, 2nd edition
- Brijlal - Heat and Thermodynamics - 1st edition
- Tofazzal Hossain - A Text Book of Heat, 2nd edition
- Beiser - Concept of Modern Physics, 6th edition

2) Supplementary Reading

- Web Source Link
- Journal papers
- Physics for Engineers – Jiaul Ahsan

3) Others:

- Lecture Outlines
- PowerPoint File
- Webpage